

BUSITEMA UNIVERSITY
FACULTY OF SCIENCE AND EDUCATION, NAGONGENRA CAMPUS
DEPARTMENT OF GEOGRAPHY
BACHELOR OF SCIENCE WITH EDUCATION DEGREE
YEAR TWO SEMESTER ONE EXAMINATION 2023/2024
GEO 2103: QUANTITATIVE AND RESEARCH METHODS IN GEOGRAPHY

DATE: 7th December 2023

TIME: 9:00 – 12:00 AM

Instructions. Attempt any **FOUR** questions including question 1

1. The data given below is for minimum temperature ($^{\circ}\text{C}$) for Kasese for the month of February in 1982 (source: UNMA).

**16.3, 18.2, 16.7, 16.3, 17.7, 17.4, 17.5, 19.1, 19.0, 18.3, 20.1, 16.7, 17.2, 17.2,
16.7, 16.7, 18.6, 19.2, 19.0, 15.4, 15.7, 18.7, 16.8, 18.0, 18.2, 14.6, 18.6, 18.0**

Calculate

- a) The range (5 marks)
 - b) The interquartile range (8 marks)
 - c) The variance (7 marks)
 - d) The standard deviation (5 marks)
2. (a) Distinguish between the following: (6 marks @)
- i. Qualitative and Quantitative data
 - ii. Descriptive and Inferential statistics
- (b) Explain the importance of a literature review in the research process (7 marks)
- (c) Describe any five types of research carried out in scientific inquiry (12 marks)
3. Critically describe the stages in the research process and how they are related to each other. (20 marks)
4. (a) Describe the sources of research problem in geographical inquiry (12 marks)
- (b) Describe five basic elements unique to geography research reports (13 marks)
5. “There was a landslide in Bududa in 2010 which claimed lives, people were advised to leave to avoid another episode. Others left while others remained. As a researcher, suggest a suitable;
- a) Title (2 mark)
 - b) Background (5 marks)
 - c) Statement of the problem (5 marks)
 - d) Objectives of the study (4 marks)
 - e) Methods of data collection (5 marks)
6. Table 1 shows the data of the weight of 40 students in a class. The measurement is to the nearest kg.
Table 1: The mass (Kg) of students

Mass (kg)	Frequency
45 – 49	2
50 – 54	4
55 – 59	7
60 – 64	10
65 – 69	4
70 – 74	6
75 – 79	7

a) Construct a cumulative frequency distribution graph showing the mass of students
(10 marks)

b) Calculate the mean, mode and median for the distribution presented in table 3
above.
(15 marks)

7. (a) Differentiate between a null and alternative hypotheses as applied in geographical research studies.
(2 marks)

(b) A Geography student tested the concentration of Total Suspended Solids (TSS) at 2 points in river Manafwa for September October and November season (SON) and obtained the data given in Table 2 below

Table 2: Suspended solid in river Manafwa

Upstream	2.5	1	2.5	6.5	4				
Downstream	11	13.5	6.5	5	8.5	11	13.5	8.5	11

a) State a null hypothesis that could be used to test the student's aim and comment on your answer.
(4 marks)

b) Test the null hypothesis in 6 (b) (i) and comment on your answer
(14 marks)

END